MICHIGAN MEDICINE
UNIVERSITY OF MICHIGAN

URINARY TRACT INFECTIONS IN ADULTS

UTI DIAGNOSTICS KEY POINTS:

- **Mental status changes in geriatric patients SHOULD NOT be worked up/treated for a UTI unless leukocytosis, hypotension, or ≥ 2 SIRS criteria are present.**

- **The presence of pyuria, nitrites or positive urine cultures are NOT diagnostic for a UTI. A urinalysis or urine culture should only be utilized to RULE OUT a UTI, not rule in.**

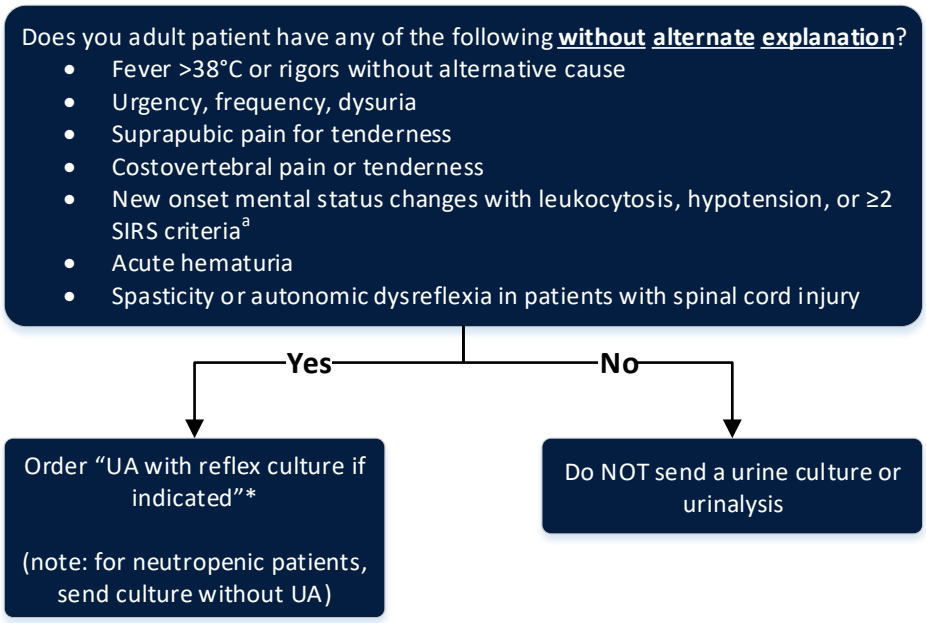
- **Except in pregnancy and prior to urologic procedures, bacteria found in a urine culture without urinary specific symptoms or symptoms of systemic infection (Asymptomatic Bacteriuria) should not be treated, regardless of presence of nitrites or pyuria, bacterial density (CFU's), or isolation of resistant organisms.**

Urinary tract infections are over-diagnosed. Several issues are often responsible for this. First, asymptomatic bacteruria (ASB) is often treated unnecessarily, and accounts for a substantial burden of unnecessary antimicrobial use and related adverse effects and resistance, as well as increased risk of future symptomatic UTI. However, ASB is so common as to be normal and expected in hospitalized patients, especially in the elderly and/or catheterized patients. **Because clinical benefit has NOT been shown in the treatment of ASB and harm has, national guidelines recommend against testing (and treating) for asymptomatic bacteriuria, except in pregnancy and prior to urologic procedures.**

Second, a positive urinalysis, in the absence of clinical signs/symptoms of a UTI, does not diagnose (or suggest) a UTI. As with ASB, pyuria, nitrites and/or the the presence of leukocyte esterase in urine is common in hospitalized patients. **Accordingly, a urinalysis can only be utilized to rule out a UTI if negative/normal.**

Finally, **given such over-diagnosis, patient histories of prior UTIs may or may not have been indicative of a true infection and should not supersede appropriate current assessment**.

As such, in the absence of signs or symptoms (see below) attributable to a urinary tract infection, patients with a positive urine culture and/or pyuria should not be treated with antibiotics irrespective of high bacterial colony count, or a multi-drug resistant organism. **The following is an effective strategy for how and when to order a urinalysis and/or urine culture.**



*: With this order, a urine culture will only be performed if a urinalysis result demonstrates pyuria or bacteriuria.

[a] SIRS Criteria: Heart rate greater than 90 bpm, respiratory rate greater than 20 breaths per minute, temperature less than 36°C, white blood count less than 4,000 cells/mm^3, temperature greater than 38°C, white blood count greater than 12,000 cells/mm^3.

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Prostatitis	Epididymitis	References

New classifications of uUTI and cUTI

Old Classifications

Uncomplicated UTI: Acute cystitis in afebrile nonpregnant premenopausal women with no diabetes and no urologic abnormalities



Acute Pyelonephritis: Acute kidney infection in women otherwise meeting the definition of uncomplicated UTI above



Complicated UTI: All other UTIs

New Classifications

Uncomplicated UTI: Infection confined to the bladder in afebrile women or men

Complicated UTI: infection beyond the bladder in women or men

- Pyelonephritis
- Febrile or bacteremic UTI
- Catheter-associated (CAUTI)
- Prostatitis* (*not covered by these guidelines)



2025 IDSA Complicated UTI guideline updates

Classification	Drug/duration	Comments
Uncomplicated UTI (bladder only) • Most outpatients • Some hospitalized	Short durations: **Nitrofurantoin** x5 days **TMP-SMX** x3 days **Beta-lactams** x 5-7 days	Beta-lactams • Generally considered inferior • Appropriate for: ○ Allergies, significant CKD, or resistance profiles • Require longer durations (5-7 days)
Complicated UTI (beyond the bladder) • Antibiotics required to have good tissue penetration	Standard duration: **TMP-SMX** x7 days **Fluoroquinolones** x5-7 days **Beta-lactams** x7 days See cUTI table below for patients where longer durations may be appropriate.	FQ • Do not treat for < 7 days in bacteremia, severe sepsis/septic shock on presentation Beta-lactams • Generally considered inferior • Appropriate for: ○ Allergies, significant CKD, or resistance profiles

Clinical Setting	Therapy	Duration	Comments
Asymptomatic Bacteriuria No symptoms of UTI (listed below) Possible UTI symptoms include (without alternative explanation): - Urgency, frequency, dysuria - Suprapubic pain or tenderness - Costovertebral pain or tenderness - Spasticity or autonomic dysreflexia in patients with spinal cord injury - New onset mental status changes with leukocytosis, hypotension, or ≥ 2 SIRS criteria[a] - Acute hematuria without alternative etiology	In most circumstances, asymptomatic bacteriuria should not be treated, regardless of pyuria, bacterial density, or isolation of resistant organisms. Treatment is recommended in the following circumstances: pregnancy and prior to urologic procedures. 1st line: **Nitrofurantoin** 100 mg PO BID (contraindicated if CrCl < 30 mL/min) Alternative: **Cephalexin*** 500 mg PO BID (avoid if high-risk allergy to penicillin or cephalosporin, see BLAES guidance) OR **Fosfomycin** 3 g PO once *Adjust dose based on renal function	Treatment of Asymptomatic Bacteriuria in PREGNANCY **Nitrofurantoin**: 5 days **Cephalexin**: 7 days **Fosfomycin**: 1 dose	• Surgical prophylaxis guidelines provide recommendations on antimicrobial prophylaxis prior to genitourinary operation • Screening for and treatment of asymptomatic bacteriuria prior to implantation of prosthetic orthopedic 2 or cardiac devices or neurosurgical procedures should not be treated • Fosfomycin should only be used to treat E. coli or E. faecalis. Fosfomycin is less effective than nitrofurantoin and should only be used if a contraindication to nitrofurantoin exists Pregnancy: • Urine culture should be sent and treatment adjusted based on susceptibilities. Follow-up urine cultures should be obtained for test of cure. • Relatively contraindicated throughout pregnancy: ○ Fluoroquinolones and doxycycline • Avoid in first 8 weeks of pregnancy: ○ TMP-SMX

Clinical Setting	Therapy (Should consider recent previous cultures)	Duration	Comments
Uncomplicated Cystitis Infection limited to the bladder in male or female patients not meeting criteria for complicated UTI or CA-UTI (see below)	Preferred **Nitrofurantoin** 100 mg PO BID (contraindicated if CrCl < 30 mL/min). 1st Alternative **TMP-SMX*** 1 DS tab PO BID 2nd Alternative **Fosfomycin** 3 g PO once OR **Cephalexin*** 1000 mg PO BID 3rd Alternative **Ciprofloxacin*** 500 mg PO BID OR **Levofloxacin*** 500 mg PO daily 4th Alternative: **Gentamicin*** 5 mg/kg IV x1 OR **Tobramycin*** 5 mg/kg IV x1 OR **Amikacin*** 15 mg/kg IV x1 (use Ideal body weight)	**Nitrofurantoin**: 5 days **TMP-SMX:** 3 days **Ciprofloxacin / levofloxacin**: 3 days **Fosfomycin**: 1 dose **Oral Beta-lactams**: 5-7 days IV → PO (PO beta-lactam, TMP-SMX, nitrofurantoin, or fluoroquinolone) 3-5 days For patients with recurrent or relapsed UTIs, or slow/delayed response to treatment, use clinician judgement for duration, as longer durations may be appropriate. In pregnancy, treatment durations should be 5-7 days.	• Fosfomycin should only be used to treat *E. coli* or *E. faecalis*. Fosfomycin is less effective than nitrofurantoin and should only be used if a contraindication to nitrofurantoin exists. • Fluoroquinolones are not recommended as 1st-line agents due to high rates of *E. coli* resistance and propensity for collateral damage (resistance, *C. difficile* infection). Use should be reserved when 1st and 2nd line options are not feasible • 3 days of IV beta-lactam may be sufficient to treat uncomplicated UTI, so if a patient is started on an IV beta-lactam and then transitioned to an oral beta-lactam (or other oral agent), shorter durations of 3-5 days are appropriate. • Oral beta-lactams are not recommended first line as they have a higher risk of treatment failure. • Single dose aminoglycoside may be a good option for patients with antibiotic-resistant cystitis with symptoms limited to the bladder in Emergency Department patients who do not otherwise have an indication for admission. Please discuss with stewardship pager (30780) with any questions or concerns. Use in caution in patients with acute kidney injury. • Renal transplant patients are beyond the scope of this guideline. Careful consideration should be given to determine if a shorter antibiotic course is appropriate for the individual patient. Pregnancy: • Urine culture should be sent and treatment adjusted based on susceptibilities. Follow-up urine cultures should be obtained for test of cure. • Relatively contraindicated throughout pregnancy: ○ Fluoroquinolones and doxycycline • Avoid in first 8 weeks of pregnancy: ○ TMP-SMX

Clinical Setting	Therapy (should take into account recent previous cultures)	Duration	Comments
Catheter-associated UTI (CAUTI): without symptoms indicating infection beyond the bladder (fevers, severe sepsis, pyelonephritis, bacteremia, urinary obstruction)	Preferred **Nitrofurantoin** 100 mg PO BID (contraindicated if CrCl < 30 mL/min) 1st Alternative **TMP-SMX*** 1 DS tab PO BID 2nd Alternative **Cephalexin*** 1000 mg PO TID 3rd Alternative **Ciprofloxacin** 500 mg PO BID OR **Levofloxacin** 500mg PO daily	7 days FQ: 5 days For patients with recurrent or relapsed UTIs, or slow/delayed response to treatment, use clinician judgement for duration, as longer durations may be appropriate.	• Asymptomatic bacteriuria with or without pyuria in catheterized patients are NOT an indication for treatment • Removal of urinary catheter is an important part of treatment. • If catheter removed and instigating symptoms have resolved, likely no infection was present and no further treatment is needed • Antimicrobial choice should be adjusted based on urine culture and susceptibility testing

Clinical Setting	Empiric Therapy (should consider recent previous cultures)	Duration	Comments
Complicated Urinary Tract Infection (Infection beyond the bladder) Including: • Febrile UTI • UTI with severe sepsis • UTI with bacteremia • Pyelonephritis • UTI with obstruction	Without known prior resistance or septic shock: **Ceftriaxone** 2 g IV daily Critically ill or septic shock: **Cefepime*** 2 g IV q8h + Vancomycin IV per pharmacy followed by oral therapy if possible* Alternative in patients with history of ESBL-producing organisms: **Meropenem** 1 g IV q8h with or without vancomycin as above followed by oral therapy if possible* **See renal dosing table for dosing recommendations in other scenarios (in particular *Pseudomonas*)** PREGNANCY: Follow recommendations as above, but note agents to avoid in comments *See appendix	Complicated UTI without bacteremia or severe sepsis/septic shock at presentation with quick clinical improvement: • If using a FQ, can use 5-day duration • Otherwise, 7 days Complicated UTI with bacteremia or severe sepsis/septic shock with quick clinical improvement: 7 days If following criteria are met: • Quinolone or TMP-SMX or IV beta-lactam used as definitive therapy For patients with relapsed complicated UTI, delayed symptom resolution, or urinary diversion, recent urologic surgery, anatomic abnormalities, use clinician judgement for duration, as longer durations may be appropriate. Renal transplant patients with complicated UTI (with infection beyond the bladder) should be treated with a minimum 14-day duration. ID consult recommended. Perinephric Abscess: Prolonged duration (see comments)	• Perinephric abscess: recommend ID and urology consult • Nitrofurantoin and fosfomycin should be avoided in complicated UTI treatment given concern for upper-tract or parenchymal involvement • Removal of urinary catheter is an important part of treatment. • If catheter removed and instigating symptoms have resolved, likely no infection was present and no further treatment is needed • Antimicrobial choice should be adjusted based on urine culture and susceptibility testing • For obstructive physiology, urology should be consulted for consideration of source control measures. Pregnancy: • Urine culture should be sent and treatment adjusted based on susceptibilities. Follow-up urine cultures should be obtained for test of cure. • Relatively contraindicated throughout pregnancy: ○ Fluoroquinolones and doxycycline • Avoid in first 8 weeks of pregnancy: ○ TMP-SMX

Clinical Setting	Therapy (should consider recent previous cultures)	Duration	Comments
Prostatitis Patients typically present with frequency, urgency, urinary incontinence, poor stream, hesitancy, fever **and** a tender, edematous prostate on exam.	Preferred: **Ciprofloxacin** 750 mg PO BID OR **Levofloxacin** 750 mg PO daily OR **TMP-SMX** 1 DS tab PO BID Alternative for patients with *E. coli* or *E. faecalis*, in consultation with ID: **Fosfomycin** 3 g PO daily x 1 week followed by 3 g PO q48h x6-12 weeks	4-6 weeks	• Antimicrobial choice should be adjusted based on urine culture and susceptibility testing
Epididymitis Primary pathogens *C. trachomatis* or *N. gonorrhoeae* Enteric organisms are more likely in men who practice insertive anal intercourse Enteric organisms in select patients only, such as after prostate biopsy, vasectomy, and other urinary tract instrumentation.	**Ceftriaxone** 500 mg** IM once + **Doxycycline** 100 mg PO BID *Men who practice insertive anal intercourse* **Ceftriaxone** 500 mg** IM once (ceftriaxone could be omitted in select cases if there is a low suspicion for gonorrhea or there is a negative NAAT for *N. gonorrhoeae*) + **Levofloxacin*** 750 mg PO daily (coverage of enteric organisms) *Adjust dose based on renal function ** If weight ≥ 150 kg, ceftriaxone dose should be 1000 mg	**Doxycycline**: 10 days **Levofloxacin**: 10 days	• Recommended tests: U/A, Urine culture, and NAATs from urine or urethral specimen for *N. gonorrhoeae* and *C. trachomatis* • **All men should be empirically treated with antibiotics that cover *C. trachomatis* and *N. gonorrhoeae*. Antimicrobial regimens should be re-evaluated based on NAAT and urine culture results.**

***Renal Dosing Recommendations**

[a] SIRS Criteria: Heart rate greater than 90 bpm, respiratory rate greater than 20 breaths per minute, temperature less than 36°C, white blood count less than 4,000 cells/mm^3, temperature greater than 38°C, white blood count greater than 12,000 cells/mm^3.

Appendix: Use of oral antibiotics for bacteremia, pyelonephritis, and other non-cystitis urinary tract infections

Oral antibiotic therapy should *preferentially* be used in patients meeting the following criteria, assuming *in vitro* susceptibility:

- Ability to take oral medications reliably
- Clinical stability
- No concerns for diminished absorption (e.g., short-gut syndrome, unavoidable drug-drug interactions)

Table 1: **Preferred** oral antibiotic dosing for bacteremia, pyelonephritis, and other complicated urinary tract infections

Antibiotic	**Bioavailability**	**Dosing (normal renal function)**
Amoxicillin (*Enterococcus* only)	70 – 80%	1000 mg PO QID
Ciprofloxacin*, €	70%	750 mg PO BID
Levofloxacin*, €	99%	750 mg PO daily
Trimethoprim-sullfamethoxazole, €	~100%	1-2 DS PO BID (2 DS preferred for bacteremia)

*Administer at least 2 hours before and 6 hours after divalent cations (calcium, magnesium, aluminum, dairy). Administration of continuous tube feeds may reduce efficacy
€ Preferred

Table 2. Oral β-lactam dosing for bacteremia, pyelonephritis, and other complicated urinary tract infections **for when fluoroquinolones and trimethoprim-sulfamethoxazole cannot be used**

Antibiotic	**Bioavailability**	**Dosing (normal renal function)**
Amoxicillin-clavulanate	70 – 80% (amoxicillin)	875 mg PO TID
Cefpodoxime	50%	400 mg PO BID

Should only be used in patients with definitive source control when applicable, resolution of clinical signs and symptoms of infection, and at least 3 days of prior intravenous therapy. Should be used ONLY in consultation with ID and when fluoroquinolones and trimethoprim-sulfamethoxazole are not options. Available pharmacokinetic / pharmacodynamic (PK/PD) data, using contemporary PK/PD targets, do not support the use of oral β-lactam agents as definitive therapy for non-cystitis urinary tract infections, with or without bacteremia. However, published clinical data and experience on the use of oral β-lactams as definitive step-down therapy do suggest that they have a role in many patients although the risk of recurrence and treatment is likely marginally higher in comparison to fluoroquinolones and trimethoprim-sulfamethoxazole. Additionally, the role of prior active therapy, definitive source control, and other patient-specific factors must be considered.

Thus, the use of oral β-lactams for non-cystitis urinary tract infections with or without bacteremia must be evaluated with patient-specific risk-benefit considerations in mind, including the potential harms of central intravenous line placement, treatment failure / recurrence, or alternative oral therapies

Last revised: 03/2026	**Antimicrobial Stewardship Subcommittee:** 10/2025	**P&T:** 09/2025

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Revision History

Date	**Revision**
2/2024	Recommendations for use of oral antibiotics
10/2025	Clarified FQ durations
03/2026	Revised Diagnostic Considerations

The recommendations in this guide are meant to serve as treatment guidelines for use at Michigan Medicine facilities. If you are an individual experiencing a medical emergency, call 911 immediately. These guidelines should not replace a provider's professional medical advice based on clinical judgment or be used in lieu of an Infectious Diseases consultation when necessary. As a result of ongoing research, practice guidelines may from time-to-time change. The authors of these guidelines have made all attempts to ensure the accuracy based on current information, however, due to ongoing research, users of these guidelines are strongly encouraged to confirm the information contained within them through an independent source